Statistical Measures

Welcome back to CS 2100!

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Exercise:

```
data = {
    'title': ['The Matrix', 'Inception', 'Interstellar', 'Blade Runner'],
    'year': [1999, 2010, 2014, 1982],
    'rating': [8.7, 8.8, 8.6, 8.1]
}
df = pd.DataFrame(data)
```

Let's:

- get the average rating
- print a sentence like `"The Matrix was released in 1999"` for each movie
- add a new movie `"Dune"` with `year=2021` and `rating=8.0`

Common Pandas operations: selecting a column

```
print(list(df['Age']))  # [10, 13]

print(df['Age'])
```

```
0    10
1    13
Name: Age, dtype: int64
```

Common Pandas operations: iterating through rows

```
for index, row in df.iterrows():
    print(f'Row number {index}:\n{row}\n')
```

```
Row number 0:
Person    Cat
Age        10
Name: 0, dtype: object

Row number 1:
Person    Elephant
Age             13
Name: 1, dtype: object
```

Common Pandas operations: iterating through rows

Each row is a dict

```
for index, row in df.iterrows():
    print(f"{row['Person']}'s age is {row['Age']}")
```

```
Cat's age is 10
Elephant's age is 13
```

Common Pandas operations: adding more rows

Create another dataframe and use `concat()` to concatenate them

```
new_rows = pd.DataFrame({
    'Person': ['Dog', 'Giraffe'],
    'Age': [3, 6]})

df = pd.concat([df, new_rows], ignore_index=True)
print(df)
```

```
     Person  Age
0       Cat   10
1  Elephant   13
2       Dog    3
3   Giraffe    6
```

Common Pandas operations: adding more columns

```
df['BFF'] = ['Cat', 'Giraffe', 'Cat', 'Elephant']
print(df)
```

```
     Person  Age       BFF
0       Cat   10       Cat
1  Elephant   13   Giraffe
2       Dog    3       Cat
3   Giraffe    6  Elephant
```

(BFF = best friend forever)

We will do more Pandas operations (like sorting and filtering) in Lecture 15.

Poll: How can I find the age of the oldest person in `df`?

1. `df(max['Age'])`
2. `max(df['Age'])`
3. `max(row['Age'] for _, row in df.iterrows())`
4. `max('Age' for _, row in df.iterrows())`

Statistical Variables

Computer programming	Statistics
a variable is a named reference to a location in the computer's memory	a variable is a characteristic that can be measured or categorized
it holds exactly one value at any time while the program runs	it represents the range of possible values across a dataset
it's designed to describe what value is currently stored and how it's used in logic / operations	it's designed to describe what values occur in a dataset
`age = 25`, `name: Optional[str] = None`, `has_bodyguard = True`	age, birth month, what time the bus arrives, whether or not it rains
variables can have many types (int, string, boolean, list, object, etc.)	variables are generally either quantitative or categorical (next section)

Poll: Is this describing a variable in the statistical or computer programming sense?

```
score = score + 5
```

1. Statistical
2. Programming

Poll: Is this describing a variable in the statistical or computer programming sense?

In our dataset, age ranges from 18 to 65

1. Statistical
2. Programming

Poll: Is this describing a variable in the statistical or computer programming sense?

age = 5, and then later, age = 6

1. Statistical
2. Programming

Poll: Is this describing a variable in the statistical or computer programming sense?

Shoe size is a variable we collected from 200 survey respondents

1. Statistical
2. Programming

Poll: Is this describing a variable in the statistical or computer programming sense?

The loop variable idx goes from 0 to 9

1. Statistical
2. Programming

Poll: Is this describing a variable in the statistical or computer programming sense?

Is zip_code a variable in this dataset?

1. Statistical
2. Programming

Statistical Variable types

Quantitative variables have numeric meaning.

Categorical variables represent labels or categories — they describe a quality or characteristic rather than a quantity.

Classic trap: Numbers that are actually categorical variables. E.g., zip code, phone numbers, student ID numbers.

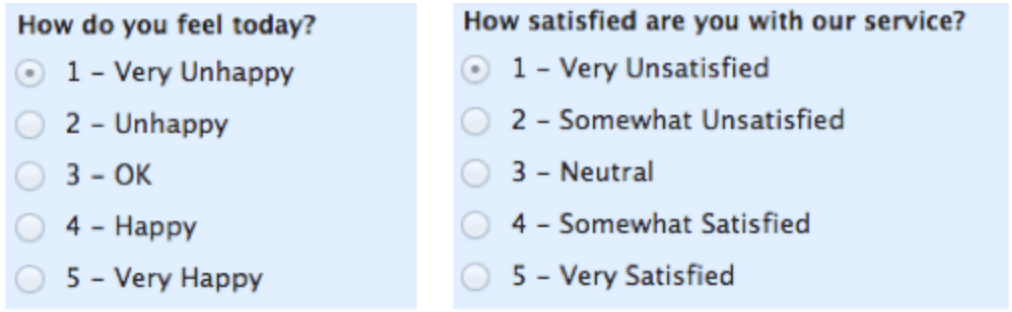
The test isn't "is it stored as a number?" but "does doing arithmetic on these values make sense?"

Special Categorical Variables

Binary: only two possible values. E.g., whether or not someone is being sarcastic, whether or not someone blinks

Ordinal: categories have an inherent order.

Ordinal variables will appear on the TRACE evaluations which you should complete at the end of the semester.



Source of that image: https://www.mymarketresearchmethods.com/types-of-data-nominal-ordinal-interval-ratio/

Quantitative Variable Types

Discrete	Continuous
can only take on specific, separate values -- there are gaps between possible values, and you can't meaningfully have a value in between two consecutive ones	can take on any value within a given range — between any two values, there's always another possible value
E.g., number of marbles, number of bugs in a program	E.g., time, temperature, distance

Poll: Should it be a continuous or discrete variable?

Number of emails in an inbox

1. Continuous
2. Discrete

Poll: Should it be a continuous or discrete variable?

Time it takes the bus to get to the airport

1. Continuous
2. Discrete

Poll: Should it be a continuous or discrete variable?

Height of a plant

1. Continuous
2. Discrete

Poll: Should it be a continuous or discrete variable?

Number of courses a student is enrolled in

1. Continuous
2. Discrete

Poll: Should it be a continuous or discrete variable?

Volume of water in a glass

1. Continuous
2. Discrete

Poll: Should it be a continuous or discrete variable?

Shoe size

1. Continuous
2. Discrete

Binning

Binning is the process of taking a quantitative variable and grouping its values into intervals or "bins."

It turns quantitative data into categorical data.

It helps with analysis, since it's easier to summarize "ages 20–29, 30–39, 40–49" than every individual age.

Here are the runtimes (in milliseconds) of 12 test cases:

12, 15, 18, 23, 26, 29, 31, 38, 42, 44, 45, 49

You decide to bin them into equal-width bins of size 10, starting at 10:

Bin A: 10 - 19	Bin B: 20–29
Bin C: 30–39	Bin D: 40–49

Poll 1: How many runtimes fall into Bin B (20–29)?

1. 2
2. 3
3. 4
4. 5

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12, 15, 18, 23, 26, 29, 31, 38, 42, 44, 45, 49

You decide to bin them into equal-width bins of size 10, starting at 10:

Bin A: 10 - 19	Bin B: 20–29
Bin C: 30–39	Bin D: 40–49

Poll 2: Which bin contains the most runtimes?

1. Bin A
2. Bin B
3. Bin C
4. Bin D

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12, 15, 18, 23, 26, 29, 31, 38, 42, 44, 45, 49

You decide to bin them into equal-width bins of size 10, starting at 10:

Bin A: 10 - 19	Bin B: 20–29
Bin C: 30–39	Bin D: 40–49

Poll 3: Instead of equal-width bins, you want 3 equal-frequency bins (so they have 4 runtimes each). What would the first bin (fastest 4 test cases) contain?

1. 12, 15, 18, 23	2. 12, 15, 18, 26
3. 12, 15, 18, 23, 26	4. All runtimes below 20

Mean and median

Mean = sum of all values ÷ number of values

Let's calculate the mean on this dataset:

Dataset: 4, 5, 6, 7, 8
Mean = 30 / 5 = 6

And this dataset:

Dataset: 4, 5, 6, 7, 50
Mean = 72 / 5 = 14.4

Does 14.4 feel like it represents this data well? (Four of the five values are way below it.)

Median = the value in the middle when data is sorted (or average of the two middle values if there's an even count)

Same dataset: 4, 5, 6, 7, 50 -> sorted, middle value = 6

Now compare: median (6) vs. mean (14.4) — median is much more representative when there's an extreme value.

Poll: What are the mean and median of this dataset? Which one better represents a "typical" value?

Dataset: 10, 12, 13, 14, 90

Poll: A company reports that thir mean salary is $120,000 per year. An employee tells you that most people there actually make around $65,000. Please propose an explanation for this discrepancy.

Variance

Here are two datasets with the same mean but very different spread:

Dataset A: 48, 49, 50, 51, 52 -> mean = 50

Dataset B: 10, 30, 50, 70, 90 -> mean = 50

How do we communicate that A is more reliably close to the mean than B?

First instinct: average distance from the mean

A natural first guess to measure "spread": average of (value – mean)

Try it on Dataset B:

Dataset B: 10, 30, 50, 70, 90

(10-50) + (30-50) + (50-50) + (70-50) + (90-50) = -40 + -20 + 0 + 20 + 40 = 0

The positive and negative **deviations** cancel out, so this naive approach always gives 0, no matter how spread out the data is. This motivates the next step...

Fix the cancellation problem: square the deviations

Since negative and positive deviations cancel, we can square each deviation first so everything is positive:

Dataset B: 10, 30, 50, 70, 90

$(\text{value} - \text{mean})^2$

For Dataset B: 1600, 400, 0, 400, 1600

Average the squared deviations -> variance

Variance = average of the squared deviations from the mean

Dataset A: 48, 49, 50, 51, 52

(1600 + 400 + 0 + 400 + 1600) / 5 = 4000 / 5 = 800

Dataset B: 10, 30, 50, 70, 90

(4,1,0,1,4)/5 = 10/5 = 2 (much smaller, so more reliably close to the mean)

Units problem -> standard deviation

If our data is in milliseconds, what are the units of variance?

milliseconds^2, squared units

Squared units can be hard to interpret intuitively. So, we take the square root.

Standard deviation = square root of variance

Poll: Dataset A: 48, 49, 50, 51, 52. Dataset B: 10, 30, 50, 70, 90. Both have a mean of 50. Which has the higher variance?

1. Dataset A
2. Dataset B
3. They're equal
4. Cannot be determined

Poll: Given the dataset 4, 8, 6, 10, 12, what are the variance and standard deviation?

Poll: Two sorting algorithms both have a mean runtime of 50ms. Algorithm A has a standard deviation of 2ms. Algorithm B has a standard deviation of 40ms. Which algorithm has a more consistent runtime?

1. Algorithm A
2. Algorithm B
3. They're equally consistent
4. Cannot be determined from this information

Poll:

1. What is your main takeaway from today?

2. What would you like to revisit next time?